

TP Report: Processing Large CSV Datasets with Pandas, Dask, and Compression

This practical work compares different techniques for processing large CSV datasets in Python. The experiment uses three approaches: Pandas with chunk processing Dask for parallel processing Reading compressed datasets using gzip The objective is to evaluate execution time, scalability, and storage optimization.

1. Dataset Generation

A synthetic dataset named **big_dataset.csv** was generated automatically using Python. The dataset contains the following columns: id device_name temperature humidity status The target file size was approximately 5 GB.

2. Processing Methods

Method	Description
Pandas Chunks	Reads the CSV file in small chunks to reduce memory usage.
Dask	Processes the dataset in parallel and handles large files efficiently.
Compression (gzip)	Uses compressed files to reduce storage size and optimize reading.

3. Code Summary

The program measures the execution time for each processing method and stores the results inside a CSV file named **comparison_results.csv**. The workflow includes: Generating a large dataset Reading the dataset using Pandas chunks Processing the dataset with Dask Compressing the dataset using gzip Comparing execution times and file sizes

4. Conclusion

This experiment demonstrates different techniques for handling large datasets in Python. Pandas chunks help reduce memory consumption, Dask improves scalability and parallel execution, and gzip compression reduces storage size efficiently. These technologies are useful in Big Data environments where datasets are too large to fit entirely into memory.